DC Armature Controlled Motor Transfer Function

The derivation of the transfer function for an armature-controlled DC motor.

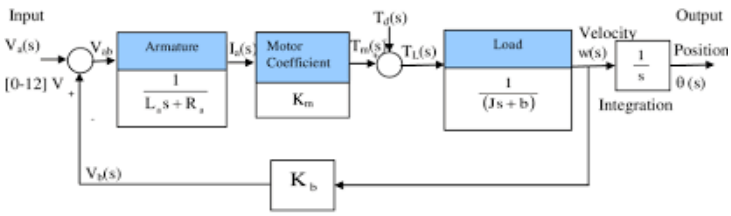

Figure 1: Armature controlled DC motor block diagram

To model this, we split the system into its electrical and mechanical components. Let's define the standard parameters:

- $v_a(t)$: Armature voltage (V)
- $i_a(t)$: Armature current (A)
- R_a: Armature resistance (Ω)
- L_a: Armature inductance (H)
- $e_b(t)$: Back electromotive force (EMF) (V)
- K_b: Back EMF constant (V·s/rad)
- K_t: Motor torque constant (N·m/A)
- $T_m(t)$: Motor torque (N·m)
- J: Equivalent moment of inertia of the rotor and load (kg·m^2)
- b: Equivalent viscous friction coefficient (N·m·s/rad)
- $\omega(t)$: Angular velocity (rad/s)
- $\theta(t)$: Angular position (rad)

1. The Differential Equations

The Electrical System (Kirchhoff's Voltage Law): The voltage applied to the armature must balance the voltage drops across the resistance, inductance, and the back EMF generated by the spinning rotor:

$$v_a(t) = R_a i_a(t) + L_a \frac{di_a(t)}{dt} + e_b(t) \tag{1}$$

Where the back EMF is proportional to the angular velocity:

$$e_b(t) = K_b \omega(t) = K_b \frac{d\theta(t)}{dt} \tag{2}$$

The Mechanical System (Newton's Second Law for Rotation): The torque generated by the motor must overcome the rotational inertia and viscous friction:

$$T_m(t) = J\frac{d\omega(t)}{dt} + b\omega(t) \tag{3}$$

Where the generated torque is proportional to the armature current:

$$T_m(t) = K_t i_a(t) \tag{4}$$

2. Laplace Transforms

Assuming zero initial conditions, we take the Laplace transform of the equations above:

1. $V_a(s) = (R_a + sL_a)I_a(s) + E_b(s)$
2. $E_b(s) = K_b s\Theta(s) = K_b\Omega(s)$
3. $T_m(s) = (Js + b)\Omega(s)$
4. $T_m(s) = K_t I_a(s)$

3. The Transfer Functions

By substituting the current and back EMF equations into the mechanical equation, we can solve for the relationship between the input voltage and the output motion.

Speed Transfer Function: If your students are designing a speed controller, the transfer function relating angular velocity $\Omega(s)$ to input voltage $V_a(s)$ is:

$$\frac{\Omega(s)}{V_a(s)} = \frac{K_t}{(R_a + sL_a)(Js + b) + K_t K_b} \tag{5}$$

Position Transfer Function: Since a capstan drive is typically used for precise positioning, we integrate the speed ($\Omega(s) = s\Theta(s)$) to get the transfer function relating angular position $\Theta(s)$ to input voltage $V_a(s)$:

$$\frac{\Theta(s)}{V_a(s)} = \frac{K_t}{s[(R_a + sL_a)(Js + b) + K_t K_b]} \tag{6}$$

(A quick side note for the students: When using standard SI units, the numerical values of the torque constant K_t and the back EMF constant K_b are identical, which simplifies the math nicely!)

4. Simplified Model (Neglecting Armature Inductance)

Assuming the motor inductance (L_a) is negligible is a common and practical simplification, especially for small brushed DC motors or coreless designs where the electrical time constant ($\tau_e = L_a/R_a$) is fractions of a millisecond.

Mathematically, setting $L_a = 0$ assumes the electrical dynamics occur instantaneously relative to the mechanical dynamics. This reduces the order of the system by one, which significantly simplifies the PID tuning process for the students.

Simplified Speed Transfer Function

Starting with our previous speed transfer function:

$$\frac{\Omega(s)}{V_a(s)} = \frac{K_t}{(R_a + sL_a)(Js + b) + K_t K_b} \tag{7}$$

By setting $L_a = 0$, the equation simplifies to:

$$\frac{\Omega(s)}{V_a(s)} = \frac{K_t}{R_a(Js + b) + K_t K_b} \tag{8}$$

Distributing the R_a and rearranging the terms gives us:

$$\frac{\Omega(s)}{V_a(s)} = \frac{K_t}{(R_a J)s + (R_a b + K_t K_b)} \tag{9}$$

To put this into the standard form for a first-order system, we divide the numerator and denominator by the constant term $(R_a b + K_t K_b)$:

$$\frac{\Omega(s)}{V_a(s)} = \frac{\frac{K_t}{R_a b + K_t K_b}}{\left(\frac{R_a J}{R_a b + K_t K_b}\right) s + 1} \tag{10}$$

This reveals the classic first-order transfer function structure:

$$\frac{\Omega(s)}{V_a(s)} = \frac{K_m}{\tau_m s + 1} \tag{11}$$

Where:

- K_m is the **Motor Gain** (rad/(V·s)): $K_m = \frac{K_t}{R_a b + K_t K_b}$
- τ_m is the **Mechanical Time Constant** (s): $\tau_m = \frac{R_a J}{R_a b + K_t K_b}$

Simplified Position Transfer Function

Because position is the integral of speed, we simply multiply the simplified speed transfer function by $\frac{1}{s}$.

The position transfer function reduces from a third-order system to a much more manageable second-order system:

$$\frac{\Theta(s)}{V_a(s)} = \frac{K_m}{s(\tau_m s + 1)} \tag{12}$$

5. Experimental System Identification

Using a simple step response is a direct way to bridge the gap between theoretical models and physical hardware. Because the system has an encoder on the motor shaft (measuring ω_m) and a position sensor on the gearbox output (measuring θ_{out}), the motor dynamics can be isolated from the gearbox kinematics.

Experiment 1: Finding the Motor Parameters (K_m and τ_m)

This experiment isolates the motor using the encoder data. A constant voltage (step input) is applied, and the motor's velocity response is recorded over time.

Procedure:

1. Ensure the capstan drive drum is disconnected or unloaded so the motor spins freely with only the inertia of the gearbox and input drum of capstan attached.
2. Command a constant step voltage V_{step} to the motor amplifier (ensuring it does not saturate the amplifier or exceed the motor's rated speed).
3. Record the motor shaft velocity $\omega_m(t)$ at a high sampling rate (e.g., 1 kHz) until the speed levels off at a constant steady-state value.

Finding the Steady-State Gain (K_m): When the motor reaches its final, constant speed, the derivative term in the differential equation becomes zero. Mathematically, as $t \to \infty$, $s \to 0$ in the transfer function:

$$\omega_{ss} = K_m \cdot V_{step} \tag{13}$$

The steady-state gain is calculated by dividing the steady-state speed (ω_{ss}) by the input voltage:

$$K_m = \frac{\omega_{ss}}{V_{step}} \tag{14}$$

Finding the Mechanical Time Constant (τ_m): The time constant τ_m dictates how fast the motor reaches steady state. For a first-order system, the response to a step input is:

$$\omega_m(t) = \omega_{ss}(1 - e^{-t/\tau_m}) \tag{15}$$

Evaluating this at $t = \tau_m$ yields:

$$\omega_m(\tau_m) = \omega_{ss}(1 - e^{-1}) \approx 0.632\omega_{ss} \tag{16}$$

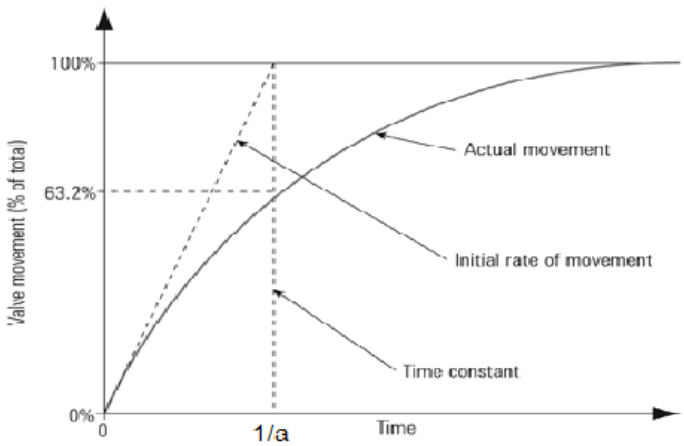

Figure 2: Step response of a first-order system showing the 63.2% time constant.

To find τ_m from the recorded data:

1. Calculate 63.2% of the final steady-state speed (ω_{ss}).
2. Find the exact timestamp t where the motor crossed that speed.
3. Subtract the time the step voltage was initially applied. The resulting duration is τ_m.

Lab Note: Students should run this at several different voltage steps, both forward and backward, and average the results to account for nonlinearities like static friction.

Experiment 2: Finding the Gear Ratio (N)

Because the gearbox output speed is not measured directly, the gear reduction ratio N is found by comparing the motor shaft position to the gearbox output position.

Procedure:

1. Start with the motor at rest. Zero both the motor encoder and the gearbox position sensor.
2. Apply a voltage to spin the motor for a few seconds, then stop.
3. Record the final position of the gearbox output shaft, θ_{out} (in radians).
4. Calculate the total distance the motor shaft traveled, θ_m, by numerically integrating the velocity data from the encoder:

$$\theta_m = \int_0^{t_{final}} \omega_m(t)dt \tag{17}$$

(Alternatively, if the encoder outputs pulse counts, convert the total pulses directly to radians).

Calculation: The gear ratio N is the ratio of the motor shaft displacement to the output shaft displacement:

$$N = \frac{\theta_m}{\theta_{out}} \tag{18}$$

The Complete System Transfer Function

With K_m, τ_m, and N determined experimentally, the full open-loop transfer function of the plant can be constructed. The relationship between the motor speed and the output position is:

$$\Theta_{out}(s) = \frac{1}{N \cdot s}\Omega_m(s) \tag{19}$$

Substituting the simplified motor model yields the final plant model:

$$\frac{\Theta_{out}(s)}{V_a(s)} = \left(\frac{1}{N \cdot s}\right)\left(\frac{K_m}{\tau_m s + 1}\right) = \frac{K_m}{Ns(\tau_m s + 1)} \tag{20}$$

6. Extracting Physical Parameters from Experimental Data

While the experimental values K_m and τ_m are sufficient for tuning a PID controller, we can also use them to back-calculate the physical parameters of the system, specifically the equivalent moment of inertia (J) and the equivalent viscous friction (b).

Recall our definitions for the motor gain and time constant from the simplified model (assuming $K_t = K_b$ in standard SI units):

$$K_m = \frac{K_t}{R_a b + K_t^2} \tag{21}$$

$$\tau_m = \frac{R_a J}{R_a b + K_t^2} \tag{22}$$

Because we have two equations and multiple unknowns, we must determine the armature resistance (R_a) and the torque constant (K_t) independently (e.g., via multimeter measurement and datasheet specifications) to solve for J and b.

Solving for Equivalent Inertia (J)

To find the inertia, we can divide the time constant by the motor gain. Notice how the shared denominator neatly cancels out:

$$\frac{\tau_m}{K_m} = \frac{\frac{R_a J}{R_a b + K_t^2}}{\frac{K_t}{R_a b + K_t^2}} = \frac{R_a J}{K_t} \tag{23}$$

Rearranging this to solve for J yields:

$$J = \frac{\tau_m K_t}{R_a K_m} \tag{24}$$

This J represents the *equivalent* inertia at the motor shaft, which includes the motor's own rotor inertia plus the reflected inertia of the capstan drum and load through the gearbox ($J = J_{motor} + J_{load}/N^2$).

Solving for Equivalent Viscous Friction (b)

To find the viscous friction, we isolate b algebraically from the K_m equation. Starting with:

$$K_m(R_a b + K_t^2) = K_t \tag{25}$$

We can distribute and rearrange the terms to solve for b:

$$R_a b = \frac{K_t}{K_m} - K_t^2 \tag{26}$$

$$b = \frac{K_t - K_m K_t^2}{R_a K_m} = \frac{K_t(1 - K_m K_t)}{R_a K_m} \tag{27}$$

Similarly, this b represents the total equivalent viscous friction at the motor shaft ($b = b_{motor} + b_{load}/N^2$).

Lab Note: If the calculated value for b is negative, it typically indicates that the theoretical assumption of purely viscous (velocity-proportional) friction is inadequate, and unmodeled non-linearities like Coulomb (static) friction are dominating the steady-state speed.